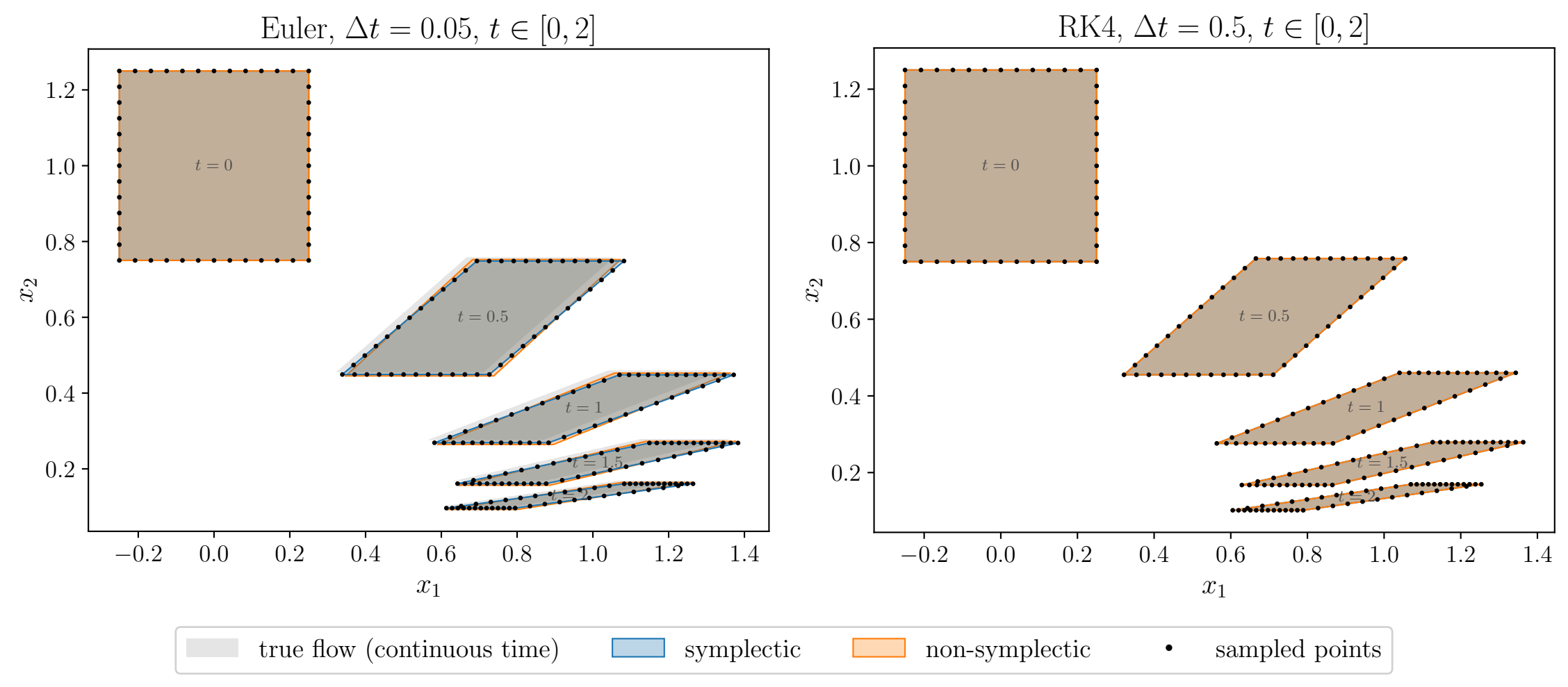

Euler, $\Delta t = 0.05$, $t \in [0, 2]$
RK4, $\Delta t = 0.5$, $t \in [0, 2]$
$t = 0$
$t = 0.5$
$t = 1$
$t = 1.5$
$t = 2$
x_1
x_2
true flow (continuous time)
symplectic
non-symplectic
sampled points